Fiscal Year:	FY 2012	**Task Last Updated:**	FY 05/04/2012
PI Name:	Fitzgerald, David Jamie Ph.D.		
Project Title:	The response of articular cartilage to microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	97239-3011	**Congressional District:**	1
Comments:	NOTE: Moved to Henry Ford Health System, Detroit, MI, mid-2015.		
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	08/01/2009	**End Date:**	07/31/2014
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:	Bion/Shuttle		
Flight Assignment:	STS-131 ; Bion M1 NOTE: End date is now 7/31/2014 per A. Chu/ARC and NSSC info (Ed., 4/30/2013)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX09AQ03G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The objective of this project is to determine whether the reduced mechanical forces associated with microgravity lead to osteoarthritis. Although the effects of microgravity on bone are well known and appear to be reversible, the effects reduced mechanical forces on the cartilage surfaces of articulating joints are less well understood. Studies in humans and animals clearly demonstrate that osteoarthritis results from abnormal (both increased and decreased) mechanical forces on cartilage. Based on these findings, we hypothesize that appropriate mechanical forces are critical for the health of articular cartilage. Following on from this, we further hypothesize that the absence of significant biomechanical forces affects that ability of the chondrocyte to maintain healthy functioning cartilage. This may lead to accelerated cartilage breakdown and osteoarthritis. There have been no studies that specifically investigate whether microgravity results in the degeneration of articular cartilage. Articular cartilage breakdown could be a major consequence of long-term space travel because, unlike bone, once cartilage degradation begins it cannot be reversed, typically leading to osteoarthritis. The mobility of an individual affected by even mid-stage osteoarthritis would be compromised leading to a failure to complete the full range of mission duties. The availability of joint tissue exposed to an extended period of microgravity is an ideal opportunity to address these hypotheses. Therefore, we have designed a focused Aim that will examine the joints of mice returned from microgravity for altered gene and protein expression and evidence of cartilage breakdown. This will be achieved using a range of robust and well-established methods including histology for joint surface integrity, immunohistochemistry using antibodies against a range of cartilage proteins and gene expression analysis tools.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We expect to determine whether cartilage degradation occurs in space. In essence, we are looking at the changes that occur under conditions of altered biomechanical forces. The effects of increased biomechanical forces have been documented for many years but the effects of greatly reduced biomechanical forces are largely unknown. We expect that these studies will provide new insights into mechano-sensitive pathways in chondrocytes, the cells that make cartilage. This in turn may lead to the development of novel therapies and treatment option for patients with osteoarthritis and other conditions where biomechanical forces on the joint are altered.
Task Progress:	Articular cartilage is located at the ends of long bones in the weight-bearing articulating joints of the knee, ankle, shoulder, hip, hand, foot and spine. It is a tissue that is precisely engineered to absorb and distribute both tensile and compressive biomechanical forces. These biomechanical forces are important for maintaining healthy articular cartilage and any deviation from the normal range of forces accelerates cartilage degradation. Alterations in the normal mechanical forces can be caused by injuries that destabilize the joint, such as meniscal cartilage or ligament damage. In addition, there is evidence from joint immobilization studies that reduced mechanical forces also lead to cartilage atrophy. Breakdown of articular cartilage initiates an irreversible cascade of degradative changes that typically leads to osteoarthritis (OA), a painful and debilitating condition that reduces mobility and has a high impact on daily activities. In humans, cartilage breakdown due to limb immobilization is rare but during long-duration spaceflight, where the normal biomechanical forces due to gravity are significantly reduced, cartilage breakdown and osteoarthritis may be a major concern. Based on these findings, we hypothesize that: The reduced biomechanical forces due to microgravity impair the ability of chondrocytes to maintain healthy articular cartilage, leading to increased cartilage breakdown. The STS-131 mission, which will expose mice to a 15-day period of microgravity, provided an ideal opportunity to test this hypothesis. We proposed to conduct cell and molecular analyses on articular cartilage of mice exposed to an extended period of microgravity, for evidence of cartilage degradation. We expect to determine whether cartilage degradation occurs as a result of microgravity. Together, the histological and immunohistochemical, and gene expression studies represent a comprehensive and robust approach that will allow us to monitor the extent and severity of cartilage breakdown. These studies will provide new insights into the role of biomechanical forces in cartilage structure and function. We have designed the study in such a way that maximizes the data generated from the small amount of tissue available. While several studies have examined the behavior of chondrocytes in simulated microgravity, there are no published studies that have investigated cartilage characteristics in mouse articular cartilage expose to microgravity aboard an orbiting vehicle. In 2012 we continued our analyses on articular cartilage and skeletal muscle tissue obtained from animals flown on STS-131. Our data indicated that 15 days exposure to microgravity does not alter proteoglycan content of articular cartilage in mouse knee joints. Longer duration exposure to spaceflight is required to adequately address the issue of whether cartilage breakdown can be detected at the tissue level. This will be examined in the 30-day Bion-M1 flight opportunity later in 2012. Gene expression analyses demonstrate that exposure to 15 days of microgravity results in a number of articular cartilage- and skeletal muscle-specific differences in gene expression. However, we found that there were six genes that were significantly increased in microgravity in both tissues. This is a significant finding because it indicates that there is a common response to microgravity between different tissues. These genes have been implicated in cellular stress response pathways. These stress response networks typically function to stop or slow down cell proliferation to give cells the opportunity to divert resources to addressing the cellular stress. This is a new and exciting finding and one that will be followed up with additional gene and protein expression experiments. These data will also allow us to frame new hypotheses for future spaceflight experiments in rodents and humans. In addition, the PI organized a workshop, entitled "NEW INSIGHTS INTO THE EFFECTS OF SPACEFLIGHT IN MUSCULOSKELETAL TISSUES," at the 2012 Orthopaedic Research Society annual meeting in San Francisco (Feb 4-7, 2012) around the STS-131 experiments: NEW INSIGHTS INTO THE EFFECTS OF SPACEFLIGHT IN MUSCULOSKELETAL TISSUES. Organizers: Jamie Fitzgerald, PhD and Stavros Thomopoulos, PhD. The response of skeletal tissues to microgravity is poorly understood. While it is recognized that bone mineral density decreases and muscles atrophy in spaceflight, very little is known about the molecular underpinnings of these effects. In addition, virtually nothing is known about the effects of microgravity on other musculoskeletal tissues such as cartilage and tendon. The speakers in this workshop all participated in a NASA mouse tissue Biospecimen Sharing Program on two Space Shuttle flights in 2010 and 2011. The purpose of this workshop is to educate the audience about the latest insights into the effects of spaceflight on bone and muscle and two previously under-investigated tissues: cartilage and tendon. The application of new cellular and molecular techniques to

	the analysis of these tissues provide new insights into the effects of microgravity on musculoskeletal tissues. Activation of Stress Response Networks in Cartilage and Skeletal Muscle During Spaceflight. Jamie Fitzgerald, PhD ; The Effects of Microgravity on the Rotator Cuff of the Shoulder. Stavros Thomopoulos, PhD ; Altered Biomechanics of Mouse Intervertebral Discs Following 15 Days of Microgravity. Jeffrey C Lotz, PhD ; Bone Loss in Microgravity is Associated with Activation of Specific Intracellular Signaling Pathways. Eduardo Almeida, PhD.
Bibliography Type:	Description: (Last Updated: 09/18/2019)
Abstracts for Journals and Proceedings	Fitzgerald J, Moscibrocki C. "Activation of Stress Response Networks in Cartilage and Skeletal Muscle During Spaceflight." 2012 Orthopaedic Research Society annual meeting, San Francisco, CA, Feb. 4-7, 2012. 2012 Orthopaedic Research Society annual meeting, San Francisco, CA, Feb. 4-7, 2012. , Feb-2012